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Paper 1: More Than a Question of Correlation? The Evidentiary Basis for Biomarker Surroga

Bibliographic Information

- **Title:** More Than a Question of Correlation? The Evidentiary Basis for Biomarker Surrogates First Used as Primary Endpoints for Cardiovascular and Metabolic Medicines in the European Union
- **Authors:** Grupstra RJ, Schoemaker V, Post AMG, van der Zande MP, Mol PGM, Bakker E, et al.
- **Journal:** Clinical Pharmacology & Therapeutics, 2026, early online
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Paper Type: GENERIC

Quick Take

This study evaluated the pre-approval scientific literature supporting 14 biomarker surrogates first used as primary endpoints in EU marketing authorizations for cardiovascular and metabolic medicines (2008–2023). It found that 13 of these surrogates were for orphan drugs, and that convincing evidence (e.g., meta-analyses) of biomarker-outcome correlation was present for only 4 of them. The authors call for better public justification of surrogate choices and for stronger validation frameworks.

Executive Summary

This study investigated the evidentiary basis for 14 novel biomarker surrogates used as primary endpoints in pivotal trials supporting initial marketing authorization of cardiovascular and metabolic medicines in the EU from 2008 to 2023. Through systematic literature searches, the authors found that for most of these biomarkers, high-quality evidence (e.g., meta-analyses) demonstrating correlation with clinical outcomes was lacking before their use in regulatory submissions. Thirteen of the 14 surrogates were used in orphan-drug settings, suggesting that regulators accept surrogates with limited validation when there is high unmet need and small trial populations. The paper emphasizes the need for better transparency in publicly available regulatory documents (EPARs) and for more rigorous validation frameworks, consistent with the recent CONSORT-Surrogate extension.

Scientific Context & Motivation

In many chronic or rare diseases, using clinical outcomes (e.g., survival or irreversible morbidity) as primary endpoints is impractical due to long latency, low event rates, or small patient populations. Biomarkers that are measured earlier can serve as surrogate endpoints if they are reasonably likely to predict clinical benefit. However, the evidentiary standard for accepting a novel biomarker surrogate remains undefined. The EMA and FDA have recognized the need for

guidance. This study provides empirical data on how new biomarker surrogates have actually been introduced in regulatory decisions, focusing on cardiovascular and metabolic diseases, and reveals gaps in correlation evidence that challenge the traditional requirement for rigorous validation.

✂ Methodological Snapshot

The authors first identified all novel biomarker surrogates by screening European Public Assessment Reports (EPARs) of cardiovascular (ATC-C) and metabolic (ATC-A) medicines approved in the EU between 2008 and 2023. A biomarker was considered ‘novel’ if it had not been used as a primary surrogate endpoint in any earlier EPAR. For each novel biomarker surrogate, systematic literature searches were conducted in PubMed and Embase to identify publications that investigated or discussed the correlation between the biomarker and the relevant clinical outcome. Studies were screened and categorized using a hierarchy of evidence framework (level 1: meta-analysis; level 2: prospective/retrospective/review; level 3: case studies). Data extraction included study design, funding, publication date, and evidence direction (positive or negative). The analysis was descriptive.

📋 Detailed Analysis

The study systematically identified 14 biomarker surrogates that were used for the first time as primary endpoints in cardiovascular and metabolic medicine approvals in the EU between 2008 and 2023. A striking finding is that 13 of 14 biomarkers were used for orphan-designated medicines, revealing a regulatory pattern of flexibility in rare diseases where traditional clinical outcomes may be impractical. The literature review showed that for only 4 of these biomarkers were there any validation studies (level 1 or 2 evidence) before approval, and the vast majority of evidence came from observational studies (level 2). Moreover, negative evidence was found for several biomarkers (e.g., serum bile acids in Alagille syndrome), yet the EMA accepted them. This suggests that regulators weigh the totality of evidence—including mechanistic plausibility, supportive secondary endpoints, and input from scientific advice—beyond formal correlation studies. The paper calls for better documentation of the rationale in public EPARs, which currently often lack explicit references. An important implication for pharmacometricians is that quantitative approaches (e.g., disease progression modeling, meta-analysis of patient-level data) could strengthen the validation of novel surrogates and provide the transparent, reproducible evidence that regulators and the public demand. The study also highlights the potential usefulness of the CONSORT-Surrogate checklist for clearer reporting in clinical trials.

Domain Context

This paper sits at the intersection of regulatory science, biomarker validation, and drug development. It directly addresses the EMA’s Regulatory Science Research Need (H1.1.1) to establish best practices for surrogate endpoint validation. For pharmacometricians, surrogate endpoints are common inputs in pharmacometric models (e.g., eGFR slope, blood pressure, LDL cholesterol), and

the validity of these surrogates underpins the reliability of model-informed drug development. The findings reinforce the importance of cross-team collaboration between pharmacometricians, statisticians, and clinicians to demonstrate robust biomarker-outcome relationships before use in pivotal trials. The paper also provides a historical context that can inform future qualification procedures and guidance development.

☒ Key Findings

- 1) Fourteen novel biomarker surrogates were identified, of which 13 (93%) were used for orphan-designated medicines. 2) For only 4 of these surrogates did pre-approval literature contain explicit validation studies (hierarchy level 1 or 2 evidence). 3) The majority (94%) of included publications were observational studies (level 2), with only 5 meta-analyses across all biomarkers. 4) Negative evidence (contradicting biomarker-outcome correlation) was identified for several biomarkers, including serum bile acids in Alagille syndrome. 5) None of the biomarkers were formally qualified by the EMA before use; but most (9/13) benefited from pre-assessment scientific advice or protocol assistance. 6) Public EPAR documentation rarely provided detailed justification with explicit references (only one case: lumasiran). 7) The only non-orphan surrogate (eGFR decrease $\geq 40\%$ in CKD) was used as part of a composite endpoint including clinical outcomes.
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Strengths & Limitations

Strengths

- Systematic, reproducible identification of novel biomarker surrogates from all initial EPARs over a 15-year period.
- Two independent researchers screened publications with a formal validation procedure to minimize bias.
- Methodology was pre-specified and search strategies were developed with a medical librarian.
- The study uses an established evidence hierarchy (Ciani et al.) to categorize publications.
- Transparent reporting of search strategies and inclusion/exclusion criteria in supplementary material.

Limitations (Acknowledged by Authors)

- Screening only abstracts may have missed relevant content if abstracts lacked explicit mention of biomarker-outcome correlation.
- Adding a third search concept for searches yielding >500 publications could have excluded relevant studies.
- Only publications before the first approval date were considered; evidence generated after approval was not included.
- Reliance on publicly available EPARs which may not contain all confidential justification.

Limitations (Expert Review)

- The cohort is limited to cardiovascular and metabolic ATC codes; results may not extrapolate to other therapeutic areas.
- The hierarchy of evidence framework (Ciani et al.) may not capture all forms of useful evidence (e.g., mechanistic models, natural history studies).
- No assessment of the post-approval validation of these surrogates using real-world data or long-term follow-up studies.
- Small number of biomarkers (14) and publications (117) limits the power of subgroup analyses.
- The literature search did not include trial registries or unpublished data, possibly missing negative findings.

Generalizability

The findings are directly relevant to cardiovascular and metabolic indications. Because most surrogates were in orphan diseases, the results may not generalize to more common diseases where larger trials and clinical outcome endpoints are feasible. However, the pattern of limited evidence may hold for other therapeutic areas, as suggested by similar findings in oncology.

Figures & Tables

- **Figure 1:** Schematic representation of the literature review methodology for each biomarker surrogate.
 - *Significance:* Provides a clear overview of the search, screening, and inclusion process.
- **Figure 2:** Flowchart showing selection of novel biomarker surrogates from EPARs (n=35 biomarkers identified, 14 novel included).
 - *Significance:* Documents the rigorous stepwise filtering process and transparency of cohort definition.
- **Figure 3:** Bar chart showing number of publications identified, screened, and included for each of the 12 biomarker surrogates with evidence.
 - *Significance:* Illustrates that few publications per biomarker were included, highlighting the limited pre-existing evidence.
- **Table 1:** Data extraction categories (variables collected from included publications).
 - *Significance:* Defines the standardized extraction framework ensuring consistency.
- **Table 2:** Characteristics of the 14 novel biomarker surrogates (indication, clinical outcome replaced, authorization type, etc.).
 - *Significance:* Core dataset showing that 13 of 14 surrogates were for orphan medicines.
- **Table 3:** Justification for biomarker surrogate choice as presented in the EPAR for each medicine.
 - *Significance:* Reveals that only one biomarker had an explicit source referenced; most lacked detailed public justification.
- **Table 4:** Characteristics of included publications (study design, hierarchy level, funding, negative evidence, etc.) for each biomarker surrogate.

- *Significance*: Shows predominance of observational studies (level 2) and presence of negative evidence for some biomarkers.
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Code & Reproducibility Assessment

No code was used; literature search strategies are provided in the supplementary material.

Supplementary Materials

Supporting Data S1 contains detailed search strategies for each biomarker in both PubMed and Embase. Figure S1 provides per-biomarker selection flowcharts. Table S1 lists the hierarchy of evidence definitions. Table S2 lists all 35 initially identified biomarkers. Table S3 includes the full data extraction table for all 117 included publications. Table S4 defines the categories of justification used in the study.

Future Directions

First, the field needs meta-analyses or pooled analyses of patient-level data to formally assess the correlation between these biomarker surrogates and clinical outcomes, particularly for orphan indications. Second, implementation of the CONSORT-Surrogate checklist should be encouraged in all trials using surrogate primary endpoints. Third, the EMA could consider publishing a publicly available list of surrogate endpoints, similar to the FDA table, with clear summaries of the evidence supporting each use. Fourth, future research should explore how secondary or composite endpoints influenced regulatory acceptance. Finally, the model-based qualification of surrogate endpoints (e.g., using survival models or joint longitudinal-survival models) should be developed and validated as a rigorous quantitative framework.

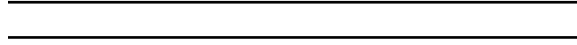
Expert Commentary

As a pharmacometrics scientist, I view this paper as a wake-up call regarding the evidence supporting many surrogate endpoints we routinely use in modeling and simulation. While mechanistic plausibility is important, regulatory decisions now rely heavily on quantitative evidence of association. The finding that meta-analytic evidence was rare suggests that our field can directly contribute by performing trial-level analyses or by building disease progression models that formally link biomarker changes to clinical outcomes. The orphan-drug context is understandable, but as models are used to extrapolate or support registration, we must ensure that the underpinning biomarker evidence is robust. This study also highlights a key role for pharmacometricians in designing studies and analyses that can generate the required level of evidence, potentially through model-based meta-analysis or multi-state modeling of natural history data.

Bottom Line

For pharmacometricians and clinical pharmacologists, this paper underscores the gap between regulatory acceptance of a surrogate endpoint and the strength of direct evidence linking it to a clinical outcome. The predominant use in orphan indications highlights a pragmatic, risk-based

acceptance that relies on mechanistic plausibility and the totality of evidence rather than formal validation. This calls for proactive generation of high-level evidence (e.g., meta-analyses) and transparent reporting in regulatory documents to support rational surrogate selection.



☒ Figures

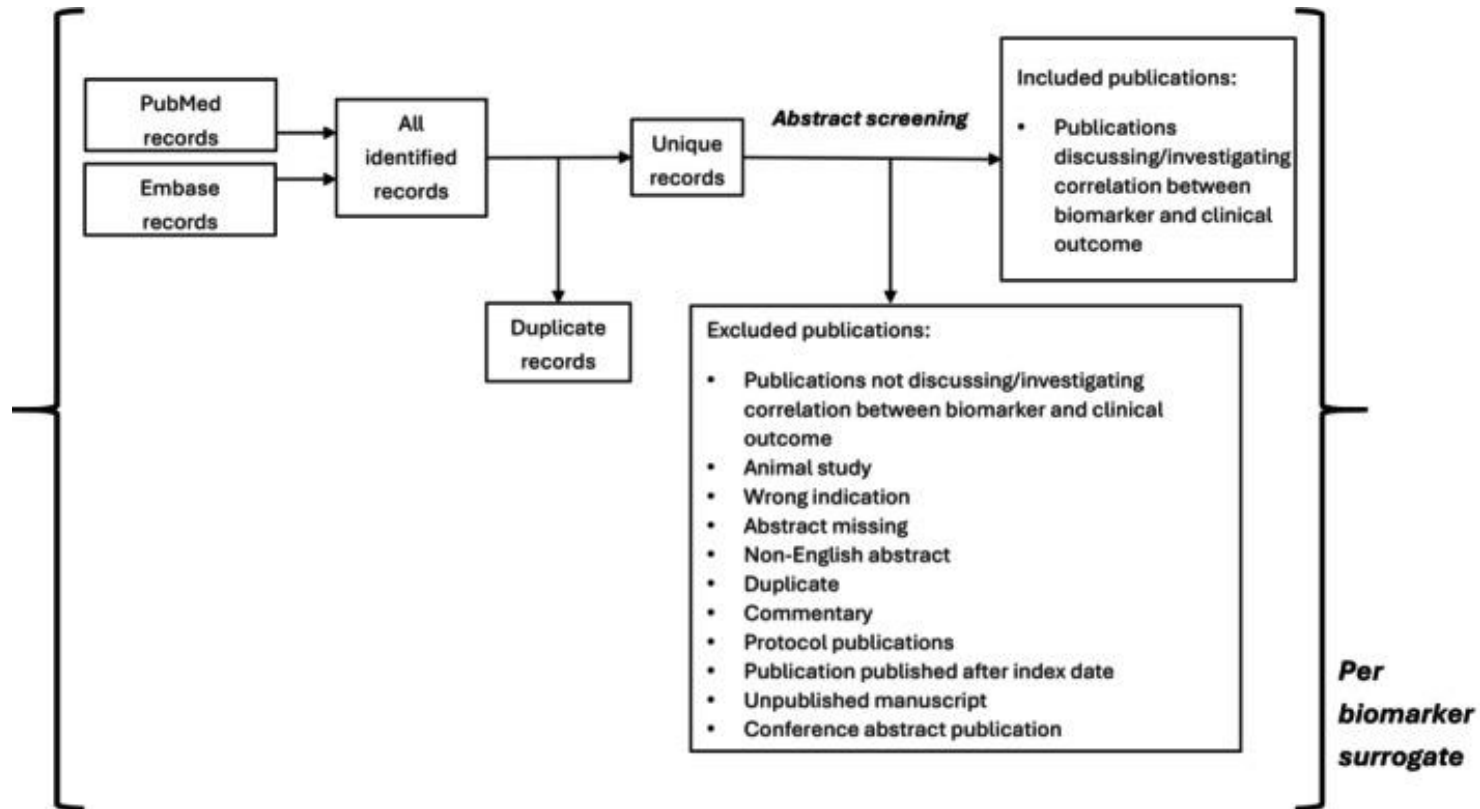


Figure 1: Schematic representation of the utilized methodology for the literature review for each identified novel biomarker surrogate.

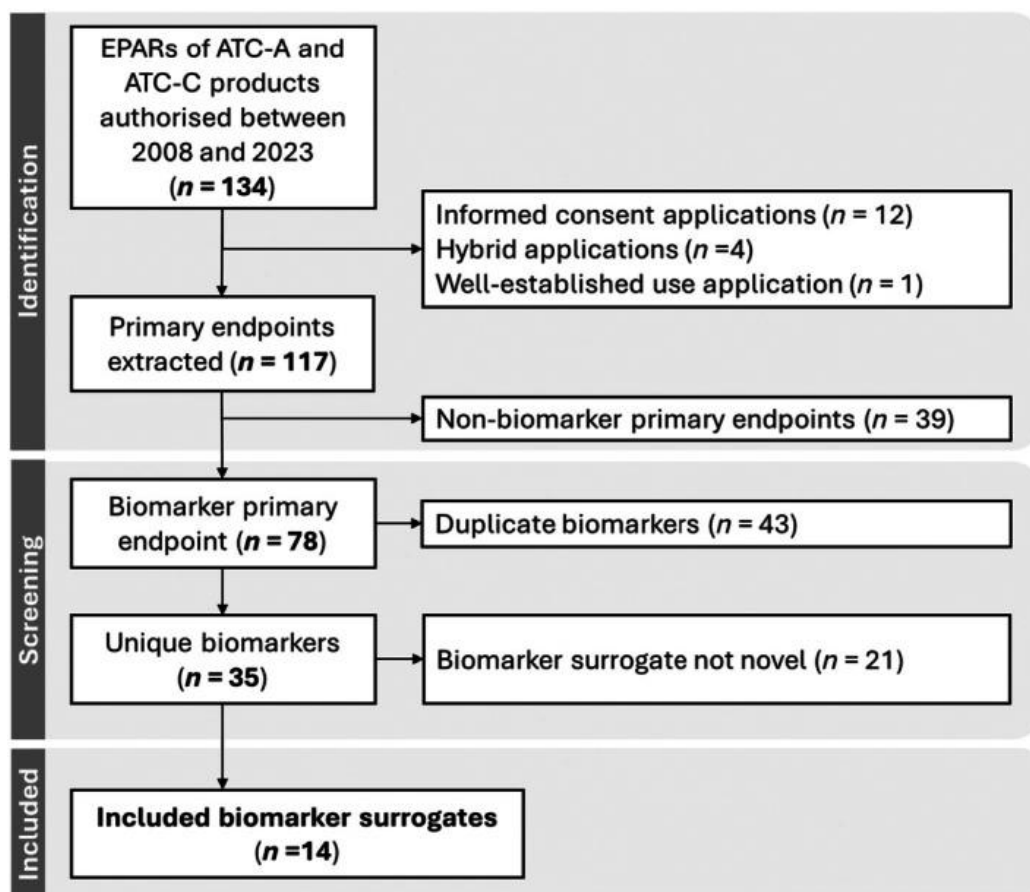


Figure 2: Flowchart for the selection of novel biomarker surrogates. EPARs were selected from the table of all EPARs from human and veterinary medicines derived from the EMA website on the J

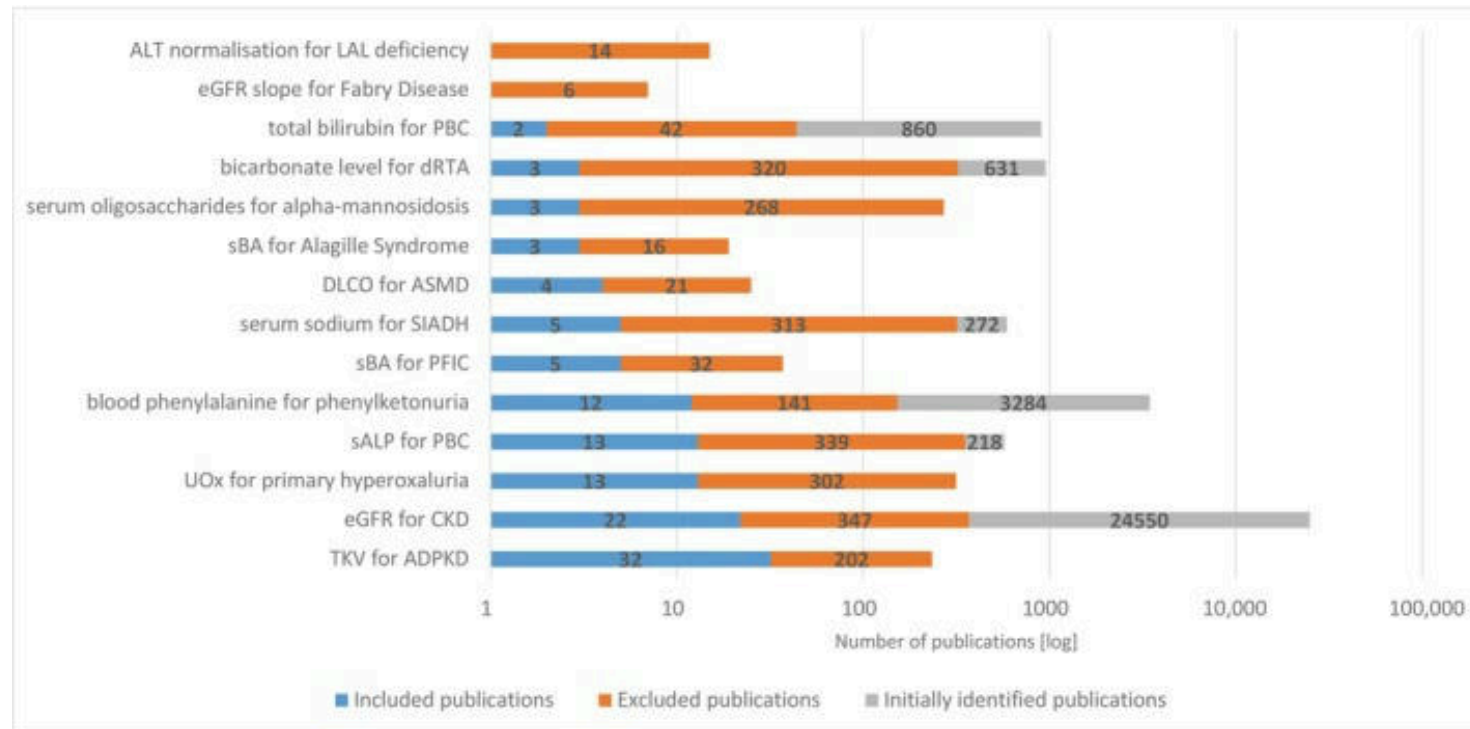


Figure 3: Overview of the identified, screened, and included publications for each biomarker surrogate. The gray bars represent the initially identified publications prior to the addition of